Automated Power Line Peeler (APLP)

Cutting Calculations: Spindle Speed, Cutting Force, Clamp Reaction,
Test 3 Analysis and Scale-Up to the SOW Case

ME 473 Capstone, Project #1 (sponsor: General Dynamics Mission Systems)
City College of New York

27 October 2026 Draft 1

1 Goal

The APLP must remove insulation from a power cable automatically. The SOW (Rev 002) asks for:

- cable outside diameter from $3/4$ to 2 in (19 to 51 mm);
- removal of up to $1/2$ in of insulation (12.7 mm, taken here as *radial*);
- a final remaining-thickness tolerance of 5 mil (0.127 mm, TBR);
- a stripped length of at least 6 in (152.4 mm);
- total power below 500 W at ≤ 50 V DC, an envelope of $6 \times 6 \times 12$ in and a mass of ≤ 20 lb.

This report sizes the cutting process for that goal. It answers four questions: (1) what spindle speed the cutter should run at, (2) what cutting force a blade would produce, (3) what clamp (reaction) force the Fusion v3 model needs so the cut does not twist, slide or bend the cable, and (4) how the bench result of Test 3 scales to the worst SOW case.

2 Reference data

2.1 SOW worst case

Quantity	Value	Source
Cable OD D	50.8 mm (2 in)	SOW
Radial removal a_{tot}	12.7 mm (R 25.4 $\rightarrow$ 12.7)	SOW, radial reading
Strip length L	152.4 mm	SOW
Remaining-thickness tolerance	± 0.127 mm	SOW (TBR)
Volume removed V	$\pi(25.4^2 - 12.7^2)\,152.4 = 231{,}667\,\text{mm}^3$	computed
Power limit	500 W total	SOW

2.2 Fusion model v3 geometry used

From "APLP Reference v3 – Open Saddle and Split Chucks FINAL" (review of 20 Oct 2026):

Item	Value
Frame	$152 \times 149 \times 290$ mm
Cable clamps	two self-centring 3-jaw split scroll chucks, A at $Z = 0$, B at $Z = 274$–290 mm
Clear span between chuck faces L_s	≈ 274 mm
Ring drive	open ring gear 72T, two 14T pinions, ratio $i = 72/14 = 5.143$
Cutter	$\varnothing 4$ mm burr on a $\varnothing 27$ mm right-angle spindle (axis radial)
Axial feed	carriage on $\varnothing 12$ rods, ball screw (SFU08-class, lead 2 mm assumed)
Carriage travel	≈ 163 mm

3 Assumptions

Every number below that is not measured is listed here. The ones marked **(calibrate)** should be replaced by bench data.

A1 **Specific cutting energy, sharp tool** (burr, end mill or blade) in XLPE/PE: $u_s = 0.1\,\mathrm{J/mm^3}$, which is a specific cutting force $k_c = 100\,\mathrm{N/mm^2}$. Textbook range for thermoplastics is 0.05–0.15 $\mathrm{J/mm^3}$. **(calibrate)**

A2 **Specific cutting energy, abrasive wheel**: $u_a = 1.0\,\mathrm{J/mm^3}$ (range 0.3–2). Grinding needs roughly 5–10 times the energy of cutting with a sharp edge, because grits rub and plough. **(calibrate)**

A3 Cutting toughness of PE/XLPE for knife slitting: $R = 15\,\mathrm{kJ/m^2}$ (range 5–30). **(calibrate)**

A4 Grinding force ratio $F_n/F_t = 2.5$ for the abrasive wheel; milling feed force $F_f = 0.5\,F_t$ for the burr; blade thrust $F_f = 0.5\,F_c$.

A5 Test 3 wheel: Dremel 426 reinforced cut-off wheel, $\varnothing 31.8$ mm, kerf $b = 1.1$ mm, run at $n = 15{,}000$ rpm (speed not recorded). **(confirm)**

A6 Test 3 cut depth $a = 1.0$ mm, i.e. through the jacket of the 7.21 mm cable (insulation thickness not measured). **(confirm)**

A7 Friction between chuck jaw and cable jacket $\mu = 0.3$ (printed or aluminium jaw on PE). Safety factor on slip $SF = 3$.

A8 A single chuck must hold the full load (conservative; the second chuck shares it in practice).

A9 2-inch cable build for stiffness: 500 kcmil compressed Al conductor $d_c = 20.65$ mm; all layers outside it lumped as XLPE, $E = 150$ MPa; stranded conductor taken at 10% of solid-rod bending stiffness ($E_{\mathrm{Al}} = 69$ GPa).

A10 SOW cutting plan and ring speed from `calculations/APLP-cycle-time.xlsx`: 5 rough passes of 2.48 mm plus a 0.3 mm finish pass, ring at 38.9 rpm, axial step 3 mm per stroke, peak removal rate $Q = 615\,\mathrm{mm^3/s}$ (pass 1).

A11 Device weight $W \approx 60\,\mathrm{N}$ ($\approx 14\,\mathrm{lb}$, Rev D estimate); v3 mass is not yet meaningful because all bodies are set to steel.

4 Test 3: helical cut on a 7.21 mm cable

Given

- Cable outside diameter $D = 7.21\,\mathrm{mm}$.
- Stripped length $L = 125\,\mathrm{mm}$.
- Total cut time $t = 30\,\mathrm{s}$.
- Time per revolution around the cable $t_r = 5\,\mathrm{s}$.
- Cutter: Dremel with abrasive cut-off wheel (A5), depth a (A6).
- Earlier result (15 Oct 2026): the Dremel and a lab sander both cut the jacket without melting it, with some hand force.

Find

(a) number of revolutions; (b) circumference; (c) helical pitch; (d) helix length per turn and in total; (e) feed rates; (f) wheel surface speed; (g) removal rate; (h) cutting power, cutting force and torque on the cable; (i) the force a knife blade would need for the same cut; (j) the time to clear the whole surface instead of scoring it.

Solution

(a) Revolutions.

$$N = \frac{t}{t_r} = \frac{30\,\mathrm{s}}{5\,\mathrm{s/rev}} = 6\,\mathrm{rev}, \qquad n_{\mathrm{ring}} = \frac{60}{t_r} = 12\,\mathrm{rpm}.$$

(b) Circumference.

$$C = \pi D = \pi(7.21) = 22.65\,\mathrm{mm}.$$

(c) Helical pitch (axial advance per revolution).

$$p = \frac{L}{N} = \frac{125}{6} = 20.83\,\mathrm{mm/rev}.$$

(d) Helix length. One turn of the helix is the hypotenuse of the unrolled triangle with sides C and p:

$$s = \sqrt{C^2 + p^2} = \sqrt{22.65^2 + 20.83^2} = \sqrt{513.1 + 434.0} = 30.77\,\mathrm{mm},$$

$$S = Ns = 6(30.77) = 184.65\,\mathrm{mm}, \qquad \text{helix angle } \psi = \tan^{-1}\frac{p}{C} = \tan^{-1}\frac{20.83}{22.65} = 42.6^\circ.$$

(e) Feed rates.

$$v_f = \frac{S}{t} = \frac{184.65}{30} = 6.15\,\mathrm{mm/s}\ (369\,\mathrm{mm/min}),$$

$$v_a = \frac{L}{t} = \frac{125}{30} = 4.17\,\text{mm/s}, \quad v_\theta = \frac{C}{t_r} = \frac{22.65}{5} = 4.53\,\text{mm/s}.$$

Check: $\sqrt{v_a^2 + v_\theta^2} = \sqrt{4.17^2 + 4.53^2} = 6.15\,\text{mm/s}$. ✓

(f) Wheel surface (cutting) speed.

$$v_s = \frac{\pi D_w n}{60} = \frac{\pi(0.0318)(15{,}000)}{60} = 25.0\,\text{m/s}.$$

The wheel edge moves about 4000 times faster than the feed (25,000/6.15), so the cut is a grinding process, not a shearing one.

(g) Material removal rate.

$$Q = b\,a\,v_f = (1.1)(1.0)(6.15) = 6.77\,\text{mm}^3/\text{s}, \qquad V_{\text{cut}} = b\,a\,S = 203\,\text{mm}^3.$$

(h) Power, force and torque. With $u_a = 1.0\,\text{J/mm}^3$ (A2):

$$P_c = u_a Q = (1.0)(6.77) = 6.8\,\text{W}, \qquad F_t = \frac{P_c}{v_s} = \frac{6.8}{25.0} = 0.27\,\text{N}, \qquad F_n = 2.5F_t \approx 0.7\,\text{N}.$$

Torque the cut puts on the cable:

$$T_{\text{cable}} = F_t \frac{D}{2} = 0.27\left(\frac{0.00721}{2}\right) = 9.8 \times 10^{-4}\,\text{N m}\ (\approx 1\,\text{N mm}).$$

Even at $u_a = 2\,\text{J/mm}^3$ the cut needs under 15 W of the Dremel's ≈ 130 W. The "hand force" felt on 15 Oct is therefore mostly the push needed to hold the wheel in the groove (normal force and friction), not cutting force. Weighing that push in the next test (Section 10) gives F_n directly.

(i) Same cut with a knife blade (slitting). A knife slitting to depth a creates new surface at a rate a per unit length of cut, so

$$F_{\text{blade}} \approx R\,a = (15{,}000\,\text{J/m}^2)(0.001\,\text{m}) = 15\,\text{N} \quad (5 \text{ to } 30\,\text{N}).$$

That is ~ 50 times the wheel's tangential force, but the blade's power is only $F v_f = 15(0.00615) = 0.09$ W and it makes no heat or dust.

(j) Clearing the whole surface. With a 20.83 mm pitch and a 1.1 mm kerf, Test 3 cut a helical groove over only

$$\frac{b}{p} = \frac{1.1}{20.83} = 5.3\%$$

of the surface; the strip between grooves still has to be peeled. To clear all of it the pitch must be $\le b$:

$$N_{\text{full}} = \frac{L}{b} = \frac{125}{1.1} = 114\,\text{rev}, \qquad t_{\text{full}} = 114(5) = 568\,\text{s} \approx 9.5\,\text{min}.$$

Check of the numbers shared with Test 3

Quantity	Shared value	Recomputed	Status
Revolutions	6	6.00	correct
Circumference	22.65 mm	22.65 mm	correct
Pitch	20.83 mm	20.83 mm	correct
Total helix length	184.65 mm	184.65 mm	correct (6×30.77)
Path feed rate	6.15 mm/s (369 mm/min)	6.15 mm/s	correct
Axial feed	4.17 mm/s	4.17 mm/s	correct
Circumferential feed	4.53 mm/s	4.53 mm/s	correct, at the OD

All seven numbers are right. Three statements around them need correcting:

1. The circumferential speed is correct at the outside surface. At the bottom of a 1 mm groove it is $\pi(5.21)/5 = 3.27\,\text{mm/s}$.
2. "Slow enough to let the wheel shear the XLPE" is not what happens: at 25 m/s the wheel grinds, and the 6 mm/s feed only sets how much heat goes into each millimetre of groove. Also, the jacket of a small test cable is usually PVC or PE, not XLPE; the material should be confirmed.
3. Test 3 was a helical *score* (5% of the surface), not a strip. It proves the wheel can follow a helix without melting the jacket; it does not yet show removal of a layer.

5 Desired spindle speed

5.1 Abrasive wheel

Reinforced cut-off wheels are rated to 35,000 rpm. For plastics a surface speed of 20–30 m/s cuts cleanly; higher speed raises heat and melting. For a wheel of diameter D_w:

$$n = \frac{60\,v_s}{\pi D_w}: \qquad D_w = 31.8\,\text{mm}:\ n = \frac{60(20\text{ to }30)}{\pi(0.0318)} = 12{,}000\text{ to }18{,}000\,\text{rpm}.$$

Recommendation: 15,000 rpm (about setting 5 on a Dremel 3000), lowered if the groove edges glaze.

5.2 Burr or end mill (Fusion v3 cutter)

Two limits set the speed: a surface speed of $v_c \approx 150$–$400\,\text{m/min}$ for carbide in PE, and a chip load of $f_z = 0.05$–$0.15\,\text{mm/tooth}$ so the tool makes chips instead of rubbing and melting. The SOW feed along the surface is $v_f = 103\,\text{mm/s}$ (ring 38.9 rpm on a 2 in cable). For a 2-flute cutter ($z = 2$):

$$n \geq \frac{60\,v_f}{z f_z} = \frac{60(103)}{2(0.155)} = 20{,}000\,\text{rpm},$$

$$v_c = \pi D_c n = \pi(0.004)(20{,}000)/60 = 4.2\,\text{m/s}\ (251\,\text{m/min})\text{ for }\varnothing 4, \qquad 6.3\,\text{m/s}\ (377\,\text{m/min})\text{ for }\varnothing 6.$$

Recommendation: 20,000 rpm for the burr spindle. A 2212-class 1000 KV motor at 24 V (24,000 rpm no load, $\approx$ 20,000 loaded) fits. If the spindle cannot reach it, slow the ring in proportion ($n_{\text{ring}} \propto n$) to keep $f_z \leq 0.15\,\text{mm}$.

5.3 Ring (orbit) speed

Test 3 ran at 12 rpm. The cycle-time sheet assumes 38.9 rpm (JGA25-370 at 200 rpm through 5.143:1), which is 3.2× faster. Going from the Test 3 diameter to 2 in at that ring speed raises surface feed from 4.5 to 103 mm/s (23×).

6 Cutting force if a blade is used

A blade can be used two ways:

Slitting (score the insulation, then peel). Force $\approx Ra$ (Test 3, part i):

$$\text{Test 3 cable, } a = 1\,\text{mm} : F \approx 15\,\text{N}; \qquad \text{SOW, } a = 12.7\,\text{mm in one slit} : F \approx 15{,}000(0.0127) = 190\,\text{N}.$$

Slitting leaves the insulation in place, so it only meets the SOW if a peel step follows and the cut stops $0.127\,\text{mm}$ from the conductor shield, which a slitting knife cannot hold.

Skiving (a lathe-style blade removes a ribbon of insulation as the ring turns). The chip area is depth a_p times feed per revolution f, so with $k_c = 100\,\text{N/mm}^2$ (A1):

$$F_c = k_c\, a_p\, f.$$

Case	a_p (mm)	f (mm/rev)	F_c (N)	$F_f = 0.5F_c$ (N)	$P = F_c v$ (W)
Test 3 cable, through jacket	1.0	0.4	40	20	0.2
SOW rough pass	2.48	0.4	99	50	10
SOW full depth, one pass	12.7	0.4	508	254	53
SOW rough pass, sheet feed	2.48	3.0	744	372	77

Here $v = 103\,\text{mm/s}$ for the SOW rows and $4.5\,\text{mm/s}$ for the Test 3 row. A blade uses little power but produces large forces: its cutting speed is the slow ring speed, so all the work has to come from force. Keeping F_c near $100\,\text{N}$ needs $f \approx 0.4\,\text{mm/rev}$, which means $152.4/0.4 = 381$ ring revolutions per pass, $381/38.9 = 9.8\,\text{min}$ per pass and about $49\,\text{min}$ for five passes.

7 Reaction force needed to cancel the cutting force (Fusion v3)

7.1 Free body

The cutter touches the cable at radius r (worst case $r = 25.4\,\text{mm}$, the first pass). It applies three loads to the cable:

- tangential F_t, which makes a torque $T = F_t r$ about the cable axis;
- radial F_r (feed force or grinding normal force), which bends the cable between the chucks;
- axial F_a from the carriage feed, which tries to slide the cable through the chucks.

The machine reacts the same loads: the ring drive carries T, the ball screw carries F_a, and the chucks carry all three back into the cable.

7.2 Cutting loads for the v3 burr

With $Q = 615\,\text{mm}^3/\text{s}$ (A10), $u_s = 0.1\,\text{J/mm}^3$ and the ∅4 burr at 20,000 rpm ($v_c = 4.19\,\text{m/s}$):

$$P_c = u_s Q = 0.1(615) = 61.5\,\text{W}, \qquad F_t = \frac{P_c}{v_c} = \frac{61.5}{4.19} = 14.7\,\text{N}, \qquad F_r = F_a \approx 0.5F_t = 7.3\,\text{N}.$$

Spindle torque: $T_{\text{sp}} = F_t D_c/2 = 14.7(0.002) = 0.029\,\text{N m}$.

7.3 Ring drive and ball screw

$$T_{\text{ring}} = F_t r = 14.7(0.0254) = 0.37\,\text{N m}, \qquad T_{\text{pinion}} = \frac{T_{\text{ring}}}{i\,\eta} = \frac{0.37}{5.143(0.5)} = 0.15\,\text{N m},$$

with $\eta = 0.5$ covering the open gear and the five V-roller pairs. A JGA25-370 gearmotor gives about 0.2–0.5 N m, so it is adequate but close; measure roller drag once the ring is built.

Ball screw (lead $\ell = 2$ mm, $\eta_s = 0.9$):

$$T_{\text{screw}} = \frac{F_a\,\ell}{2\pi\eta_s} = \frac{7.3(0.002)}{2\pi(0.9)} = 0.0026\,\text{N m},$$

negligible against a NEMA 17 (≈ 0.4 N m).

7.4 Chuck clamp force

The jaws hold the cable by friction. For one chuck (A8) with total jaw normal force N on radius $r_c = 25.4$ mm, slip is prevented when

$$\mu N \geq SF\sqrt{\left(\frac{T}{r_c}\right)^2 + F_a^2} \quad \Longrightarrow \quad N = \frac{SF}{\mu}\sqrt{\left(\frac{T}{r_c}\right)^2 + F_a^2}.$$

Burr: $T/r_c = 0.37/0.0254 = 14.7$ N,

$$N = \frac{3}{0.3}\sqrt{14.7^2 + 7.3^2} = 10(16.4) = 164\,\text{N total}, \quad 55\,\text{N per jaw}.$$

Skiving blade, SOW rough pass ($F_c = 99$ N, $F_a = 50$ N):

$$N = 10\sqrt{99^2 + 50^2} = 1{,}110\,\text{N}, \quad 370\,\text{N per jaw}; \qquad T_{\text{ring}} = 99(0.0254) = 2.5\,\text{N m}.$$

Skiving blade, full depth ($F_c = 508$ N): $N = 10\sqrt{508^2 + 254^2} = 5{,}680$ N (1,890 N per jaw), $T_{\text{ring}} = 12.9$ N m. That is not practical for printed split scrolls.

Device weight case. If the APLP hangs on a vertical or steep cable, the chucks also carry its weight in friction:

$$N = \frac{SF\,W}{\mu} = \frac{3(60)}{0.3} = 600\,\text{N}.$$

For the burr this case, not the cut, sets the clamp force. On a horizontal cable the weight presses the jaws radially and needs no friction.

Jaw pressure. At 370 N on a 10×16 mm jaw pad, $p = 370/160 = 2.3$ MPa, below the ≈ 10 MPa compressive yield of PE but enough to leave marks and to creep. Use wide, curved pads.

7.5 Cable bending between the chucks

The radial force bends the cable over the 274 mm span. Bending stiffness (A9):

$$EI = 150 \times 10^6\,\frac{\pi}{64}(0.0508^4 - 0.02065^4) + 0.1(69 \times 10^9)\frac{\pi}{64}(0.02065^4) = 47.7 + 61.6 = 109\,\text{N m}^2.$$

Mid-span deflection, between the clamped-clamped and pinned-pinned limits:

$$\delta = \frac{FL_s^3}{192\,EI} \text{ to } \frac{FL_s^3}{48\,EI}, \qquad L_s^3 = 0.0206\,\text{m}^3.$$

Cutter	F_r (N)	δ (mm)	vs. 0.127 mm tolerance
Burr (rough)	7.3	0.007 – 0.029	OK
Abrasive wheel, 300 W	30	0.03 – 0.12	marginal
Skiving blade, rough	50	0.05 – 0.20	fails if pinned
Skiving blade, full depth	254	0.25 – 1.0	fails

Deflection grows as the insulation is removed (the stripped section is less stiff), so the finish pass should be light and the depth should be referenced by a follower shoe riding on the cable next to the cutter, not by the screw position alone.

8 Scale-up from Test 3 to the SOW case

8.1 Scaling of removal rate

The SOW case removes 231,667 mm^3 against 203 mm^3 in Test 3 (1,140×). At the planned peak rate,

$$\frac{Q_{\mathrm{SOW}}}{Q_{\mathrm{T3}}} = \frac{615}{6.77} = 91.$$

Power and force scale with Q for a given process, so Test 3 calibrates u (Section 10) and the SOW numbers follow from it. Required removal rate and power for a given cutting time:

Cutting time	$Q = V/t$ (mm^3/s)	× Test 3	P, sharp tool (W)	P, abrasive (W)
5 min	772	114	77	772 (over limit)
10 min	386	57	39	386
15 min	257	38	26	257

8.2 Comparison of cutters on the SOW case

	Burr ∅4 (v3)	Abrasive ∅32	Blade, rough	Blade, full
Spindle speed (rpm)	20,000	15,000	none	none
Cutting power (W)	62	300 (limited)	10	53
Tangential force (N)	14.7	12	99	508
Radial / axial force (N)	7.3	30	50	254
Ring torque (N m)	0.37	0.30	2.5	12.9
Clamp force, one chuck (N)	164	120*	1,110	5,680
Cable deflection (mm)	≤ 0.03	≤ 0.12	≤ 0.20	≤ 1.0
Cutting time (min)	9.8	12.9	49	9.8
Total cycle† (min)	10.7	≈ 14	≈ 50	≈ 11
Verdict	best fit	power-bound	too slow	force-bound

* torque term only; the radial grinding force adds bending, not slip. † cutting plus 50 s setup and retract (cycle-time sheet). Abrasive cutting time assumes 300 W of the 500 W budget goes into the cut: $t = u_a V/P = 1.0(231{,}667)/300 = 772$ s. A ∅32 wheel also reaches only about 11 mm past its hub, so a ∅38 wheel is needed for the full 12.7 mm.

8.3 Power budget (burr, worst case)

Item	Power (W)
Spindle: 61.5 W cut /0.75 motor and ESC efficiency	82
Ring gearmotor (0.37 N m at 38.9 rpm, $\eta = 0.3$)	5
Ball-screw stepper (holding current)	10
Chuck motors, sensors, ESP32	10
Total	$\approx$ 107 (limit 500)

The burr leaves a 4$\times$ margin on power. The abrasive wheel uses most of the budget for a slower cycle. The blade is low on power but its forces exceed what the v3 chucks and ring can react, and holding 5 mil under 100–500 N is unlikely.

8.4 Conclusion

For the SOW case the v3 layout with a burr or 2-flute plastics end mill at about 20,000 rpm is the consistent choice: about 62 W into the cut, 15 N of cutting force, 0.37 N m on the ring, about 160 N of clamp force per chuck (or 600 N if the unit hangs on a vertical cable), and a cycle of about 11 min. These numbers rest on A1; if Test 4 shows u_s is twice the assumed value, power and force double and the cycle stays the same, still inside the 500 W limit.

9 Open questions for GDMS (needed to close the SOW numbers)

1. What cable is the 2 in OD case (voltage class, conductor size and material, insulation material XLPE/TR-XLPE/EPR, shields, jacket)? Can GDMS share a datasheet or a short scrap sample?
2. Is 1/2 in of insulation radial (per side) or diametral?
3. Which layers are removed: jacket only, jacket plus insulation shield, or down to the conductor shield? What remains under the 5 mil tolerance (remaining thickness, or cut depth), and is it bilateral?
4. What cycle time is acceptable? This sets whether an abrasive process is allowed (Section 8).
5. Is a helical score plus manual or automatic peel acceptable, or must all material be removed by the machine?
6. Is the line energized or de-energized during use, and what orientation is it in (horizontal span, vertical riser, any)? This sets the clamp force (160 vs. 600 N).
7. What clamp marks are acceptable on the jacket outside the strip zone?
8. Operating temperature range? XLPE stiffness and cutting energy change strongly with temperature.
9. Power source: battery on the device, or a $\leq$ 50 V DC supply cable?
10. Must chips and dust be contained (adverse environments, conductive dust)?
11. Is 6 in the minimum strip length, and what is the maximum?
12. Which tolerances in the SOW (envelope, mass, 5 mil) can be relaxed, since the SOW says they "can be changed based on team input"?

10 Next bench test (Test 4) to replace the assumptions

1. Record the Dremel speed setting, wheel or bit part number, and the cable's jacket and insulation thickness (calipers on a cut end).
2. Plug the Dremel into a plug-in wattmeter. Cutting power $P_c = P_{\text{loaded}} - P_{\text{idle}}$.
3. Measure groove width and depth after the cut; compute $Q = b\,a\,v_f$ and then $u = P_c/Q$ (replaces A1 or A2).
4. Rest the cable on a kitchen or postal scale while cutting; the reading is the normal force F_n (checks A4).
5. Repeat at two feeds (for example 5 and 2.5 s per revolution) and two speeds to see where the groove starts to melt.
6. Repeat with a ∅4–6 burr or 2-flute end mill, since that is the v3 cutter.